

Comparing two proportions

$$N = \frac{Z_{\alpha/2} \sqrt{2p(1-p)} + Z_{1-\beta} \sqrt{p_1(1-p_1)p_2}}{(p_1 - p_2)^2}$$

$Z_{\alpha/2}$ -1.96 for alpha 0.05

$Z_{1-\beta}$ - 0.84 for power 0.80

p1 and **p2** - proportion of event of interest (outcome) for group I and group II

p - (p1+p2) / 2

Comparing two means

$$N = \frac{(r + 1)(Z_{\alpha/2} + Z_{1-\beta})^2 \sigma^2}{rd^2}$$

r = n1/n2 - the ratio of sample size

σ - pooled standard deviation

d - difference of means of 2 groups

$Z_{1-\beta}$ - 0.84 for power 0.80

$Z_{\alpha/2}$ -1.96 for alpha 0.05

Descriptive studies

$$N = \frac{Z_{\alpha/2}^2 \times P \times (1 - p) \times D}{E^2}$$

N - sample size

P - prevalence or proportion of event

E - precision (or margin of error) with which a researcher want to measure something

D - design effect reflects the sampling design used in the survey type of study. This is 1 for simple random sampling and higher values (usually 1 to 2) for other designs such as stratified, systematic, cluster random sampling

$Z_{\alpha/2}$ - 1.96 for alpha 0.05